CaS-DETR: Cascaded Sparsity-to-Capacity DETR for Roadside Small-Object Detection

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Abstract. Roadside camera images often contain large background regions and many distant targets, and the number of objects can vary substantially across scenes. In DETR-like models, dense token processing can weaken the representation of small objects because informative target tokens are mixed with a large number of low-value background tokens, while a fixed processing pattern may not adapt well to scenes with different object densities. In this paper, we propose **CaS-DETR**, a DETR-based detector for roadside small-object detection. CaS-DETR follows a cascaded sparsity-to-capacity design that combines two sequential components: a **Context-Aware Semantic Sparsity** module that adaptively retains informative coarse tokens and filters low-value background tokens, and an **Object-MoE Decoder** that then refines object queries on the retained representation. This sparsity-to-capacity design reduces redundant background tokens before global interaction and allocates decoder capacity to harder object queries. Experimental results show that, compared with DEIM, CaS-DETR improves $mAP^{50:95}$ from **0.694** to **0.696** on the DAIR-V2X-I dataset and from **0.692** to **0.698** on the UA-DETRAC dataset. The gains are larger on small-object metrics, including **+7.5%** in AP_S^{50} on DAIR-V2X-I and **+1.0%** in AP_S^{50} on UA-DETRAC. Ablation and scene-density analysis further show that the adaptive token-retention strategy in Context-Aware Semantic Sparsity helps adjust the number of retained tokens according to the object density of roadside scenes.

Keywords: Roadside perception · Small-object detection · DETR · Token sparsification · Mixture of experts

1 Introduction

Roadside perception is an important component of V2X systems because infrastructure cameras provide a wider view than on-board sensors and can help detect traffic participants at intersections, on urban roads, and in other complex traffic environments [8, 26, 28, 32]. However, roadside camera images often contain large background regions together with many distant or visually small objects. As a result, detectors need to preserve weak small-object cues while reducing interference from redundant background regions. DETR-like detectors are attractive for this setting because they provide end-to-end detection and reduce the reliance on hand-crafted post-processing [2, 5, 10, 15, 16, 33, 35, 37]. However, dense token processing can be suboptimal in roadside scenes. At coarse feature scales, many tokens correspond to sky, road surface, buildings, or other low-value background regions, while distant objects may occupy only a few feature cells. Since object density can vary substantially across roadside scenes, uniformly processing all tokens may introduce redundant background interaction in sparse scenes and make small-object cues harder to preserve in crowded scenes.

To address this issue, we propose **CaS-DETR**, a cascaded sparsity-to-capacity DETR built on DETR with Improved Matching (DEIM) [5]. Our design follows a cascaded pipeline. First, the **Context-Aware Semantic Sparsity (CASS)** module suppresses redundant background tokens before encoder interaction. Second, the **Object Mixture-of-Experts (Object-MoE) Decoder** operates on the retained representation and allocates additional conditional capacity to object-query refinement. This cascaded design is intentional: the first stage reduces background-heavy redundancy, and the second stage refines harder queries on the resulting sparse representation. The overall pipeline remains simple: it first sparsifies the encoder input and then increases decoder capacity where it is most useful.

This paper focuses on a practical design for roadside small-object detection, combining scene-adaptive token sparsification with decoder-side expert refinement. Empirically, the proposed design improves both overall performance and small-object detection, with larger gains observed on small-object metrics.

Our contributions are summarized as follows:

- We propose **CaS-DETR**, a cascaded sparsity-to-capacity DETR that combines encoder-side token sparsification with decoder-side expert refinement for roadside small-object detection.
- We introduce the **CASS** module, which uses soft Gaussian supervision and a dynamic keep-ratio to preserve small-object neighborhoods while suppressing redundant S_5 background tokens.
- We introduce an **Object-MoE Decoder** that routes object queries to top-K experts, improving decoder-side refinement of roadside small-object queries.
- Experiments on the DAIR-V2X-I dataset and the UA-DETRAC dataset show consistent gains over strong baselines, especially on small-object metrics, demonstrating the effectiveness of adaptive token retention under varying roadside scene density.

2 Related Work

2.1 Roadside Small-Object Detection

Roadside perception datasets have stimulated research on infrastructure-side perception. Representative datasets include DAIR-V2X [32], Rope3D [31], and IPS300+ [24]. Based on these datasets, existing studies have mainly improved roadside perception from two perspectives: enhancing robustness under different roadside scenarios and improving the detection of small or occluded traffic targets.

The first line of work addresses scenario generalization and view-aware modeling [29, 30]. The second line focuses on 2D small-object detection. FPPNet [12] uses fixed-perspective priors to suppress background interference, while RSO-YOLO [25] improves the detection of small and occluded traffic targets. Other multi-scale fusion pipelines [36] also improve the handling of distant and occluded targets. Robust 2D detection is often used as a prior for more complex roadside monocular 3D tasks [14]. Although these methods improve feature extraction or fusion, most of them still process spatial regions densely, which can preserve low-value background information and limit adaptation to scenes with different object densities.

2.2 Efficient Detection and DETR Variants

Object detection has evolved from classic two-stage and one-stage frameworks [9, 11, 18, 23] to efficient YOLO variants [6, 7, 22] and DETR-style models such as RT-DETR, D-FINE, and DEIM [5, 16, 35]. The DETR line is attractive for roadside perception because it offers end-to-end optimization and strong global modeling. Compared with conventional dense prediction, DETR-style representations provide a clearer interface for token-level filtering before encoder interaction. However, many DETR variants still perform dense interaction over feature tokens, and existing sparse DETR designs are not specifically tailored to background-heavy roadside scenes with many small objects.

2.3 Dynamic Tokens and Mixture of Experts

Token sparsification has been widely studied in vision transformers [1, 3, 13, 17, 21]. Most of these methods aim to reduce redundant computation in general visual recognition tasks. In object detection, Sparse-DETR [20] explores query sparsity, but it is not specifically designed for roadside small-object detection. Existing sparsification methods therefore leave open the problem of how to reduce redundant coarse feature tokens in background-heavy roadside scenes while preserving local context around small objects. Mixture-of-Experts (MoE) models [4, 19] provide a way to increase model capacity through sparse activation. However, how to combine feature-token sparsification with query-level expert refinement remains less explored for roadside small-object detection.

3 Methodology

3.1 Overview

We build CaS-DETR on the DEIM framework [5]. Given an input image, the backbone and neck produce multi-scale features $\{S_3, S_4, S_5\}$. Our method modifies the baseline with a cascaded two-stage design. In the first stage, CASS predicts token scores on the S_5 feature map and retains an adaptive subset of tokens before encoder interaction. In the second stage, each decoder feed-forward Network (FFN) is replaced with a top-k MoE block for object-query refinement on the retained representation. Fig. 1 illustrates the full pipeline.

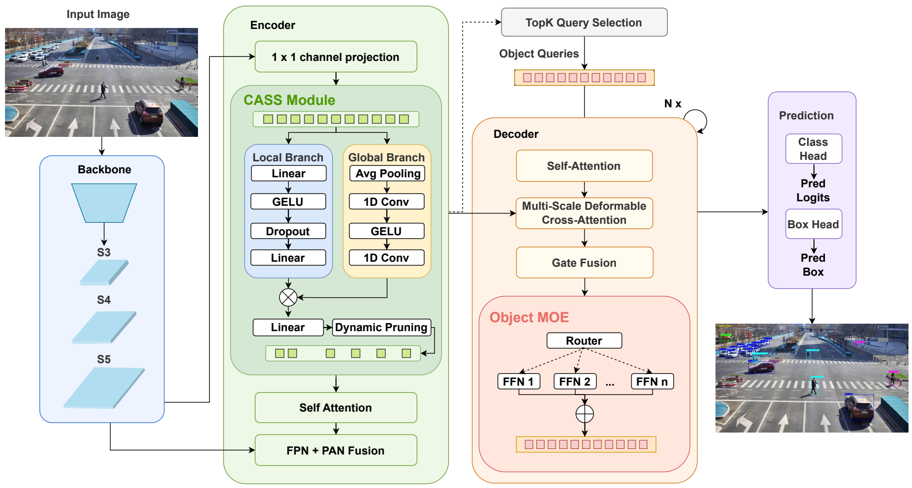

Fig. 1. Overview of CaS-DETR. Relative to DEIM, CaS-DETR introduces a cascaded two-stage design: (i) **CASS** first scores S_5 tokens and retains an adaptive subset before encoder interaction; and (ii) the decoder then applies a top-k **Object-MoE** module for query refinement on the retained representation.

Backbone. Given an input image $\mathbf{I}$, the backbone produces multi-scale features $\{S_3, S_4, S_5\}$. We apply token scoring to S_5 because it contains the highest-level semantic abstraction while remaining compact enough for global interaction. This choice follows the baseline design and allows CaS-DETR to intervene at the stage where coarse semantic redundancy is most obvious.

Encoder. Instead of treating token pruning only as a way to reduce arithmetic operations, we use it as a representation-level filter. In background-heavy roadside scenes, small targets can be weakened when large numbers of coarse background tokens participate equally in encoder interaction. CASS aims to suppress these low-value tokens while preserving a target-adjacent neighborhood through soft supervision.

Decoder. After sparsification, the retained representation is passed to the decoder. We replace the standard FFN in each decoder layer with a sparse MoE block so that decoder capacity is allocated to object-query refinement. This sparsity-to-capacity design reallocates computation toward more informative encoder tokens and harder object queries while keeping the overall complexity close to the baseline.

3.2 Context-Aware Semantic Sparsity

Token scoring. Let $\mathbf{X} \in \mathbb{R}^{N \times C}$ denote the flattened S_5 tokens. CASS predicts a score $s_j \in [0, 1]$ for each token:

$$s_j = \sigma(f_{\text{cass}}(\mathbf{x}_j)), \tag{1}$$

where f_{cass} is a lightweight two-branch scorer. In our implementation, it combines a local branch and a global branch to estimate token importance from complementary cues, and $\sigma(\cdot)$ is the sigmoid function. Specifically, the local branch uses a multi-layer perceptron to evaluate individual token features, while the global branch employs average pooling and 1D convolutions to aggregate sequence-level context, ensuring the token scores are aware of the overall scene density.

Soft supervision with a semantic halo. A hard token label generated only from box interiors is often too brittle for small objects. We therefore assign a soft target y_j to token j using an expanded box and an isotropic Gaussian decay around each ground-truth instance:

$$y_j = \max_m \left(\mathbb{1}[\mathbf{p}_j \in B_m^+] \exp\Big(- \frac{\|\mathbf{p}_j - \mathbf{p}_m\|_2^2}{2r_m^2} \Big) \right), \tag{2}$$

where $\mathbf{p}_j = (u_j, v_j)$ is the token center, $\mathbf{p}_m = (u_m, v_m)$ is the center of ground-truth box m, B_m^+ denotes an expanded support region around box m controlled by γ; in our experiments, $\gamma = 0.8$. The Gaussian radius r_m is computed from the expanded box size in an isotropic manner. This soft target preserves a local neighborhood around each object rather than supervising only the box center. In practice, this is useful because small roadside targets are often represented by very few coarse cells.

Dynamic keep-ratio. Instead of using a fixed keep-ratio for all images, we adapt the token budget to the input scene:

$$r_{\text{keep}} = \text{clip}(r_{\text{base}} + (1 - r_{\text{base}})\, \bar{s}\, \alpha_{\text{comp}}, r_{\text{base}}, 1)\,, \quad \bar{s} = \frac{1}{N} \sum_{j=1}^{N} s_j, \tag{3}$$

where α_{comp} is a density compensation factor used to scale the mean token score $\bar{s}$. We then keep the top $\lceil r_{\text{keep}} N \rceil$ tokens according to their predicted scores s_j. In practice, the base keep-ratio r_{base} is set to 0.3 on DAIR-V2X-I and 0.5 on UA-DETRAC, reflecting the different scene statistics of the two datasets. This formulation ensures that sparse scenes retain fewer tokens, while denser and more complex scenes automatically receive a larger token budget.

Sparse encoder interaction and feature restoration. Encoder self-attention is applied only to the retained S_5 tokens. After this sparse interaction, the resulting token features are scattered back to their original spatial locations, while unselected positions are filled with zeros for compatibility with the subsequent multi-scale pipeline. This design preserves spatial alignment with the baseline while avoiding unnecessary encoder computation on low-value background tokens.

3.3 Object-MoE Decoder

We replace the standard FFN in each decoder layer with a sparse MoE block. Given object query $\mathbf{q}_i$, the router predicts expert weights

$$\mathbf{g}_i = \text{softmax}(W_r \mathbf{q}_i). \tag{4}$$

It then selects the top-K experts and renormalizes their weights. The output is

$$\mathbf{y}_i = \sum_{e \in \mathcal{A}_i} \tilde{g}_{i,e} \, E_e(\mathbf{q}_i), \tag{5}$$

where $\mathcal{A}_i = \mathrm{TopK}(\mathbf{g}_i, K)$, $K = 2$, and $E_e(\cdot)$ denotes expert e. We use $E = 4$ experts in all experiments. This design increases conditional decoder capacity while keeping most computations sparse.

The decoder-side placement is intentional. Unlike a large-scale MoE applied to dense spatial tokens, our MoE operates on object queries after encoder-side sparsification. This keeps the query interface of the baseline unchanged while allowing different experts to specialize in different query refinement patterns.

3.4 Training Objective

The total loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{deim}} + \lambda_{\text{cass}} \mathcal{L}_{\text{cass}} + \lambda_{\text{moe}} \mathcal{L}_{\text{bal}}, \tag{6}$$

where $\mathcal{L}_{\text{deim}}$ is the original detection loss from DEIM [5]. We supervise token scores with token-level Varifocal Loss [34]:

$$\mathcal{L}_{\text{cass}} = \frac{1}{N} \sum_{j=1}^{N} \mathrm{VFL}(s_j, y_j). \tag{7}$$

To avoid routing collapse, we also apply a standard MoE load-balancing term

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^{E} \bar{p}_e \bar{f}_e, \tag{8}$$

where $\bar{p}_e$ is the average router probability for expert e and $\bar{f}_e$ is the fraction of queries assigned to expert e.

Baseline detection loss. We inherit the original DEIM detection objective, including its classification and box regression components. This choice keeps the training protocol close to the baseline and makes it easier to attribute changes in performance to the proposed sparse encoder and MoE decoder rather than to a new matching or regression recipe.

CASS supervision. The token-level Varifocal Loss encourages the scoring network to assign higher scores to foreground tokens and their nearby context. Compared with hard binary supervision, the soft target in Eq. (2) provides a smoother optimization signal around small objects and reduces the risk of over-pruning near object boundaries.

MoE load balancing. The auxiliary load-balancing term prevents a small number of experts from receiving most queries. In practice, this is important because the object-query distribution can be skewed in roadside scenes, especially when the number of small targets varies substantially across images.

4 Experiments

4.1 Experimental Setup

Datasets and metrics. We evaluate on DAIR-V2X-I [32] and UA-DETRAC [27]. For DAIR-V2X-I, we use 7,058 images and split them into 5,042/604/1,412 for training/validation/test. Following the original setting, the rare barrowlist category is merged into `cyclist`. For UA-DETRAC, we merge the official subsets and re-split the data by sequence into 5,007/692/1,259 images to avoid leakage from adjacent frames. We report $mAP^{50:95}$, mAP^{50}, $AP_S^{50:95}$, and AP_S^{50}, and use parameters and giga floating-point operations (GFLOPs) as complexity metrics, since these provide a more consistent basis for comparison across different hardware and deployment settings.

DAIR-V2X-I contains high-resolution roadside views with considerable scale variation and many small objects at intersections, making it a suitable benchmark for the proposed scene-adaptive sparsification strategy. UA-DETRAC provides a complementary surveillance-style roadside setting. Evaluating on both datasets lets us assess whether the same sparsity-to-capacity design remains useful across two different roadside distributions.

Implementation details. All models are initialized from official pre-trained weights and trained for 100 epochs. Unless otherwise noted, CaS-DETR inherits the optimizer and augmentation settings of DEIM, and the baselines are trained with the recommended configurations in their official implementations. The CASS reduction ratio is 4. We set $\lambda_{\text{cass}} = 0.05$, $\lambda_{\text{moe}} = 0.005$, $\alpha_{\text{comp}} = 1.0$, and use four experts with top-2 routing. Router noise with standard deviation 0.1 is added during training. For reproducibility, all baselines are trained and evaluated under the same label mapping and the same train/validation/test splits as CaS-DETR.

The spatial supervision targets are generated using the Gaussian decay strategy in Eq. (2). The base keep-ratio is set to 0.3 on DAIR-V2X-I and 0.5 on UA-DETRAC, reflecting the different scene statistics in the two datasets.

4.2 Main Results

Table 1 shows that CaS-DETR achieves the best metrics among the compared methods in this evaluation setting. Relative to DEIM, the overall $mAP^{50:95}$ gains are modest (**+0.2%** on DAIR-V2X-I and **+0.6%** on UA-DETRAC), while the improvements are more pronounced on small-object metrics. On DAIR-V2X-I, AP_S^{50} increases from **0.586** to **0.661**. On UA-DETRAC, the corresponding gain is from **0.389** to **0.399**. Relative to DEIM, the full model keeps the overall GFLOPs close to the baseline while delivering clearer gains on small-object metrics.

Table 1. Comparison with representative detectors on DAIR-V2X-I and UA-DETRAC. All methods are trained and evaluated under the same data split and label space used in this paper. Model complexity is reported in terms of parameters and GFLOPs. Best results are shown in bold.

Method	Params (M)	GFLOPs	DAIR-V2X-I				UA-DETRAC			
			$mAP^{50:95}$	mAP^{50}	$AP_S^{50:95}$	AP_S^{50}	$mAP^{50:95}$	mAP^{50}	$AP_S^{50:95}$	AP_S^{50}
Faster R-CNN	41.3	268.2	0.661	0.887	0.353	0.566	0.494	0.683	0.090	0.160
YOLOv5-S	9.1	23.8	0.636	0.851	0.274	0.500	0.514	0.655	0.153	0.245
YOLOv5-M	25.1	64.0	0.657	0.860	0.303	0.515	0.612	0.775	0.178	0.294
YOLOX-S	8.9	26.5	0.623	0.874	0.309	0.570	0.623	0.790	0.190	0.307
YOLOX-M	25.2	73.3	0.660	0.889	0.330	0.583	0.619	0.777	0.166	0.267
YOLOv8-S	11.1	28.4	0.643	0.855	0.296	0.534	0.577	0.730	0.155	0.243
YOLOv8-M	25.8	78.7	0.658	0.860	0.298	0.480	0.588	0.736	0.164	0.264
YOLO12-S	9.2	23.2	0.640	0.863	0.235	0.413	0.603	0.766	0.179	0.286
YOLO12-M	20.1	70.4	0.663	0.863	0.303	0.494	0.604	0.762	0.179	0.286
RT-DETRv2	20.0	61.2	0.683	0.905	0.361	0.588	0.634	0.792	0.230	0.360
D-FINE	10.2	25.3	0.695	0.910	0.324	0.542	0.680	0.844	0.223	0.350
DEIM	10.2	25.3	0.694	0.909	0.363	0.586	0.692	0.863	0.248	0.389
CaS-DETR	15.0	25.7	**0.696**	**0.911**	**0.391**	**0.661**	**0.698**	**0.867**	**0.251**	**0.399**

The same pattern appears across both benchmarks. On DAIR-V2X-I, where wide-view intersection scenes produce many small and distant targets, the improvement is most visible on the small-object metrics. On UA-DETRAC, the gain is smaller but remains consistent. This result is useful because the two datasets differ in viewpoint, scene layout, and category distribution, yet the same sparsity-to-capacity design remains beneficial.

These results suggest that CaS-DETR improves detection quality under comparable inference GFLOPs. CASS suppresses low-value background tokens before encoder interaction, while the MoE decoder provides additional capacity for object-query refinement through sparse routing. As a result, the model achieves a favorable accuracy-complexity trade-off, although the MoE decoder also increases parameter count. Overall, the experiments are consistent with this design goal, while not by themselves establishing a unique causal mechanism.

4.3 Ablation Study

To ensure a controlled and interpretable evaluation, all ablation experiments are conducted on DAIR-V2X-I (Table 2), which provides diverse traffic densities and small-object scenarios. **CASS only** slightly reduces GFLOPs and improves small-object metrics, suggesting that token sparsification is useful even without extra decoder capacity. This indicates that the scoring-and-selection stage is effective even on its own, reducing redundant computation while preserving or slightly improving the target representation.

MoE only increases small-object accuracy more strongly, but also increases compute and does not improve mAP^{50}. This suggests that additional decoder capacity is helpful, but not sufficient by itself to yield the most balanced result under dense encoder processing. In other words, more capacity alone does not

Table 2. Component ablation on DAIR-V2X-I. CASS improves token efficiency, MoE improves small-object refinement, and their combination gives the best balance between overall accuracy, small-object performance, and computation.

Model	CASS	MoE	GFLOPs	$mAP^{50:95}$	mAP^{50}	$AP_S^{50:95}$	AP_S^{50}
DEIM (Baseline)			25.3	0.694	0.909	0.363	0.586
DEIM + CASS	✓		24.8	0.693	0.909	0.367	0.596
DEIM + MoE		✓	26.3	0.695	0.908	**0.394**	0.621
CaS-DETR	✓	✓	25.7	**0.696**	**0.911**	0.391	**0.661**

automatically translate into cleaner query refinement when the encoder still exposes the decoder to many irrelevant background tokens.

The full model combines these two effects and achieves the best $mAP^{50:95}$, mAP^{50}, and AP_S^{50}, while remaining close to the MoE-only variant on $AP_S^{50:95}$. Compared with the MoE-only variant, CaS-DETR keeps the extra conditional capacity while reducing GFLOPs from 26.3 to 25.7 and substantially improving AP_S^{50} from **0.621** to **0.661**. Overall, the ablation supports complementarity between sparsification and decoder capacity, showing that the two modules work better together than in isolation.

4.4 Scene-Conditional Redundancy and Dynamic Sparsity

We further analyze scene-conditional redundancy and dynamic sparsity on DAIR-V2X-I (Table 3 and Fig. 2), leveraging its wide spectrum of object densities to clearly demonstrate the effect of the adaptive keep-ratio. The results suggest that a single fixed keep-ratio is not optimal across scenes with different densities. The average keep-ratio selected by CaS-DETR increases monotonically from **0.569** to **0.634** and **0.738** as scene density rises, indicating that the CASS module indeed allocates more tokens to denser scenes.

Fig. 2 reveals two useful observations. First, semantic redundancy is real, but not uniform across scenes. In low-density scenes, increasing the keep-ratio from 0.7 to 1.0 brings negligible overall improvement, with $mAP^{50:95}$ remaining at **0.691** and mAP^{50} changing only marginally from **0.893** to **0.894**. This suggests that many additional tokens retained in sparse scenes are semantically redundant. In medium-density scenes, the moderately sparse policy $r = 0.7$ even surpasses

Table 3. Scene-complexity split on DAIR-V2X-I and the average keep-ratio selected by CaS-DETR. The split is formed by sorting images according to the number of annotated objects and partitioning them into three equal-sized subsets.

Subset	Images	Avg. Objects	Avg. Keep-Ratio
Low-density	471	34.1	0.569
Medium-density	471	44.9	0.634
High-density	470	60.1	0.738

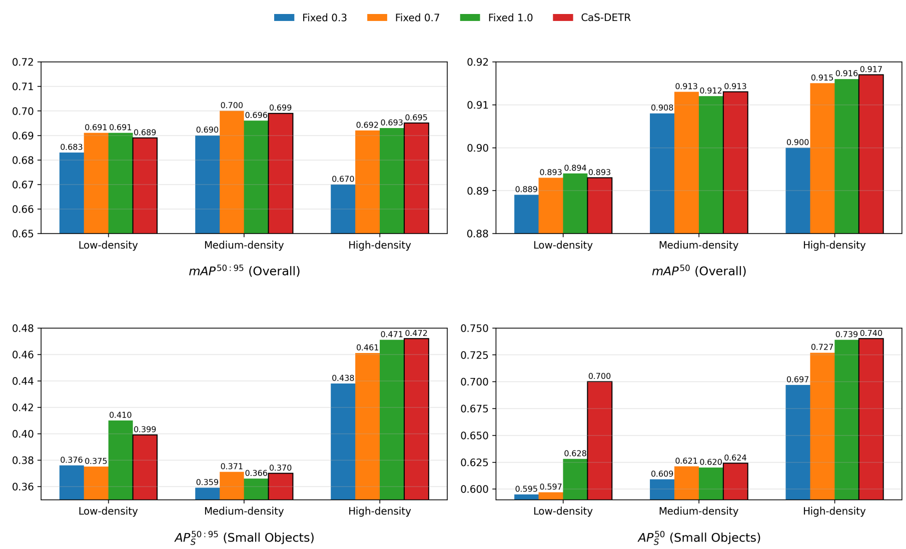

Fig. 2. Scene-conditioned comparison of fixed and dynamic keep ratios on DAIR-V2X-I. Dynamic sparsity allocates fewer tokens in sparse scenes and more tokens in crowded scenes, matching or exceeding the best fixed policy on most small-object metrics.

the fully dense setting $r = 1.0$ across all four metrics, showing that excessive background tokens can indeed be harmful. In contrast, in high-density scenes, dense retention becomes more favorable, with $r = 1.0$ outperforming $r = 0.7$, especially on small-object detection (AP_S^{50}: **0.739** vs. **0.727**). Meanwhile, the aggressively sparse baseline $r = 0.3$ performs worst across all subsets, confirming that redundancy does not imply arbitrary pruning is safe.

Second, dynamic CaS-DETR tracks the scene-dependent optimum better than any fixed policy. In low-density scenes, it preserves only **56.9%** of the tokens on average, yet achieves the best AP_S^{50} of **0.700**. In medium-density scenes, with an average keep-ratio of **0.634**, it remains essentially on par with the best fixed policy in overall accuracy while still obtaining the best mAP^{50} and AP_S^{50}. In high-density scenes, its keep-ratio further rises to **0.738**, and CaS-DETR surpasses all fixed baselines on all four metrics, reaching **0.695** $mAP^{50:95}$, **0.917** mAP^{50}, **0.472** $AP_S^{50:95}$, and **0.740** AP_S^{50}.

These results show that the advantage of CaS-DETR does not come from a single universally optimal keep-ratio. Instead, its strength lies in dynamically adapting the token budget to scene complexity: pruning more aggressively when redundancy is high, and retaining more tokens when dense object interactions require richer contextual support. This scene-aware sparsity is one of the clearest pieces of evidence supporting the overall design.

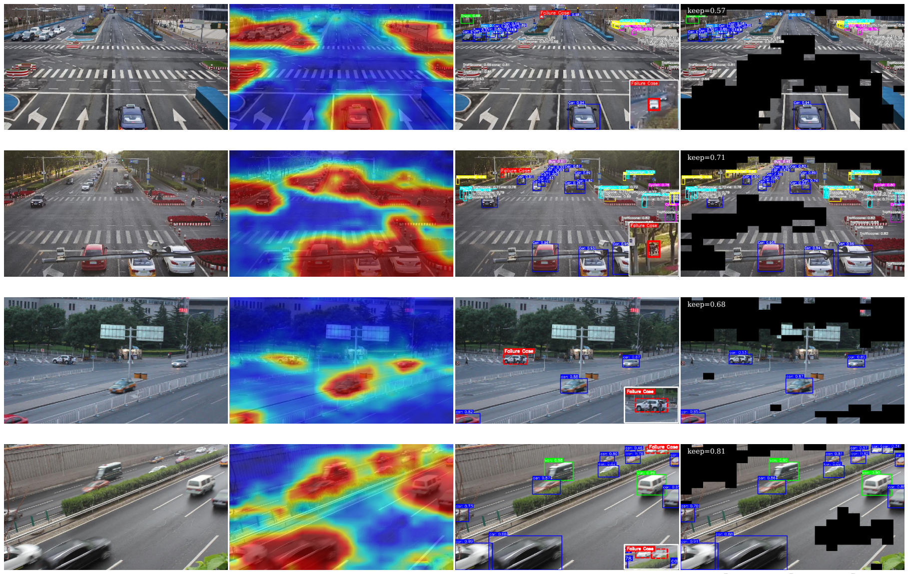

Fig. 3. Qualitative analysis of CaS-DETR on representative roadside scenes. From left to right: original image, S_5 importance map, DEIM baseline prediction, and CaS-DETR S_5 token mask with prediction. The importance maps and token masks show that CASS suppresses large background regions while preserving target neighborhoods. Compared with DEIM, CaS-DETR yields more reliable predictions on challenging small-object cases. Red boxes indicate representative baseline misses that are recovered by CaS-DETR.

4.5 Qualitative Analysis

Fig. 3 visualizes the learned token sparsification and the corresponding detection results on representative roadside scenes. The S_5 importance maps show that high-response regions are not uniformly distributed over the image. Instead, they are concentrated around semantically meaningful traffic areas, such as vehicles, road participants, lane regions, and object-adjacent context, while large sky, road-surface, and roadside background regions receive lower responses. This indicates that CASS learns to distinguish informative target-related regions from redundant background regions at the coarse semantic level.

The token masks further show the scene-adaptive behavior of CASS. In relatively sparse scenes, CaS-DETR keeps a smaller portion of S_5 tokens, such as the example with a keep-ratio of 0.57, where most large background regions are removed while the visible vehicles and their surrounding context are still retained. In more crowded or structurally complex scenes, the keep-ratio increases, for example to 0.71 and 0.81, allowing the model to preserve more tokens around multiple vehicles and dense traffic participants. This qualitative pattern is consistent with the scene-density analysis in Sect. 4.4: sparse scenes contain more removable redundancy, whereas dense scenes require a richer retained representation.

The prediction results also illustrate how the sparsity-to-capacity design benefits small-object detection. Compared with the DEIM baseline, CaS-DETR recovers several challenging distant or small targets marked by red boxes. These recovered cases are typically located in regions where the importance maps assign relatively high responses and where the token masks preserve target neighborhoods. This suggests that CASS does not simply prune tokens for efficiency, but filters background interference while maintaining useful local context for small objects. Together with the Object-MoE decoder, this leads to more reliable query refinement and fewer missed detections in challenging roadside scenes. These qualitative observations provide visual support for the small-object improvements reported in Table 1.

5 Conclusion

We presented CaS-DETR, a cascaded sparsity-to-capacity detector for roadside small-object detection. The method couples scene-adaptive token sparsification in the encoder with MoE-based query refinement in the decoder. By addressing key challenges in roadside perception, CaS-DETR filters low-value tokens from large background regions before encoder interaction, preserves target-adjacent context for distant and visually small objects, and adjusts the retained token budget according to scene-level object density. Experiments on DAIR-V2X-I and UA-DETRAC demonstrate consistent overall gains and more pronounced improvements on small-object metrics. Additional ablation studies, scene-density analysis, and visualization further show that adaptive token retention mitigates background interference in sparse scenes while maintaining richer contextual support in crowded scenes. Overall, CaS-DETR improves roadside small-object detection under comparable inference GFLOPs by cascading background-aware sparse encoding with capacity-adaptive query refinement.

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